LAPORAN PRAKTIKUM MOBILE PROGRAMMING

Pengaturan Layout Row dan Column di Flutter

Nama : Faril Isra Albiansyah

NIM : 240605110087

Kelas : Mobile Programming Praktikum A

Tanggal : 10 – 09 – 2026

JURUSAN TEKNIK INFORMATIKA

FAKULTAS SAINS DAN TEKNOLOGI

UNIVERSITAS ISLAM NEGERI MAULANA MALIK IBRAHIM

MALANG

GANJIL 2025/2026

I. Tujuan

- Memahami konsep *main axis* (sumbu utama) dan *cross axis* (sumbu silang) pada *widget* Row dan Column.
- Mengetahui fungsi dan penggunaan properti ukuran ruang seperti MainAxisSize.max dan MainAxisSize.min.
- Mempelajari nilai-nilai pada properti MainAxisAlignment untuk mendistribusikan ruang kosong antar *widget*.
- Mempelajari properti CrossAxisAlignment untuk mengatur penempatan posisi *widget* pada sumbu silang.
- Mengimplementasikan seluruh properti tata letak tersebut untuk merancang antarmuka pengguna (UI) secara terstruktur dalam kode Flutter.

II. Langkah Kerja

A. Persiapan Project

1. Membuat project Flutter baru dengan nama alignment.
2. Mengatur judul aplikasi sesuai dengan praktikum.
3. Menyiapkan struktur dasar aplikasi menggunakan MaterialApp, Scaffold, dan AppBar.

B. Pembuatan Custom Widget

1. Membuat widget KotakBiruJempol menggunakan Container.
2. Mengatur ukuran, warna, border, border radius, dan icon pada widget.
3. Membuat widget KotakBiruJempolKecil dengan ukuran yang lebih kecil.

C. Percobaan MainAxisSize

1. Menerapkan MainAxisSize.max pada Row dan Column.
2. Mengubah menjadi MainAxisSize.min.
3. Mengamati perubahan penggunaan ruang pada masing-masing layout.

D. Percobaan MainAxisAlignment

1. Menerapkan MainAxisAlignment.center.
2. Menguji start dan end.

3. Menguji spaceBetween, spaceAround, dan spaceEvenly.
4. Mengamati perubahan posisi widget pada Row dan Column.

E. Percobaan CrossAxisAlignment

1. Menerapkan CrossAxisAlignment.center.
2. Menguji start, end, dan stretch.
3. Menerapkan CrossAxisAlignment.baseline sesuai kebutuhan percobaan.
4. Mengamati perubahan posisi widget pada sumbu silang.

F. Pembuatan Tampilan Tugas

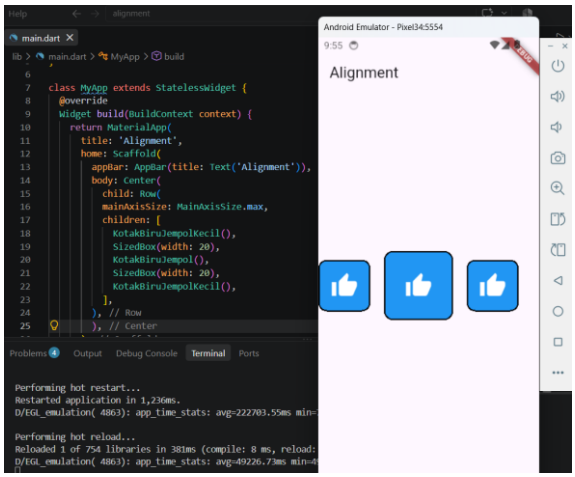
1. Membuat tampilan informasi cuaca menggunakan Column.
2. Menambahkan nama kota dan informasi suhu.
3. Menambahkan informasi prakiraan cuaca menggunakan Row.
4. Menggunakan icon sunny, cloudy_snowing, dan cloud.
5. Mengatur pembagian posisi icon menggunakan MainAxisAlignment.spaceEvenly.

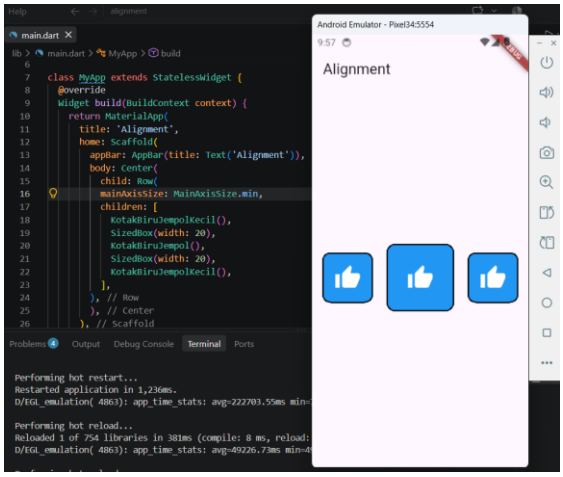
III. Screenshot Hasil

A. Percobaan MainAxisSize

1. Kode Program dan Hasil

Masukkan screenshot kode ketika melakukan percobaan MainAxisSize.max dan MainAxisSize.min.



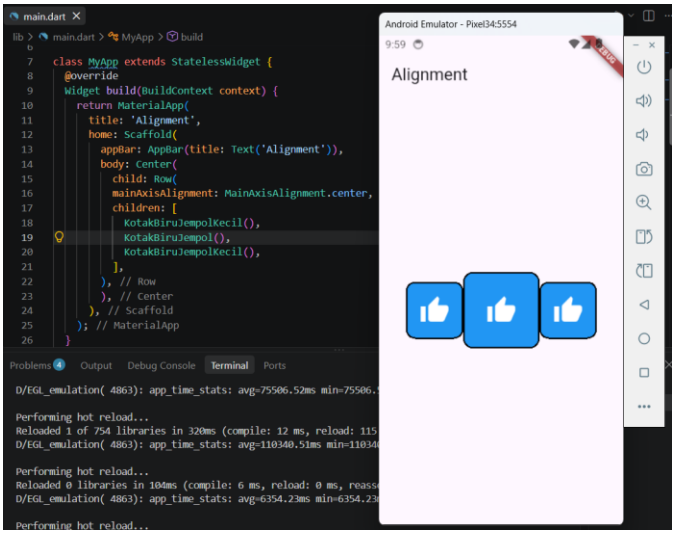


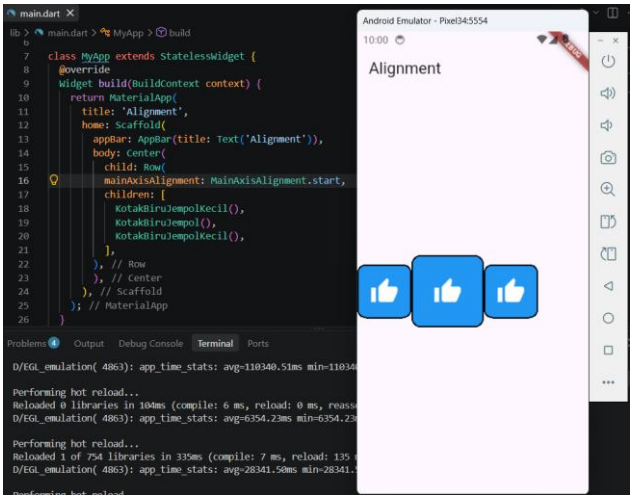
Keterangan Implementasi MainAxisSize pada Row dan Column untuk mengatur penggunaan ruang pada layout.

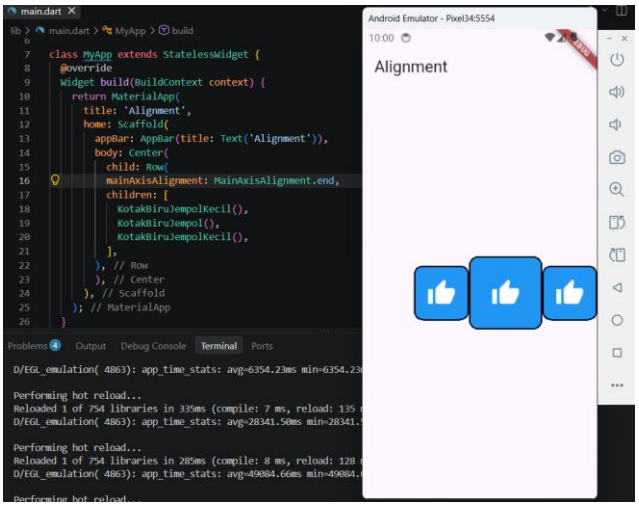
B. Percobaan MainAxisAlignment

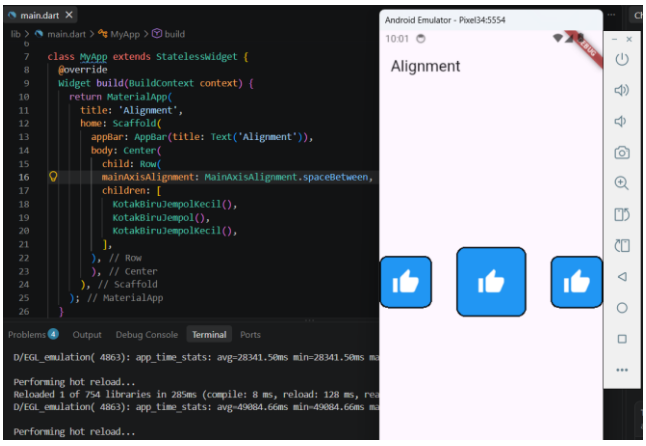
1. Kode Program dan Hasil

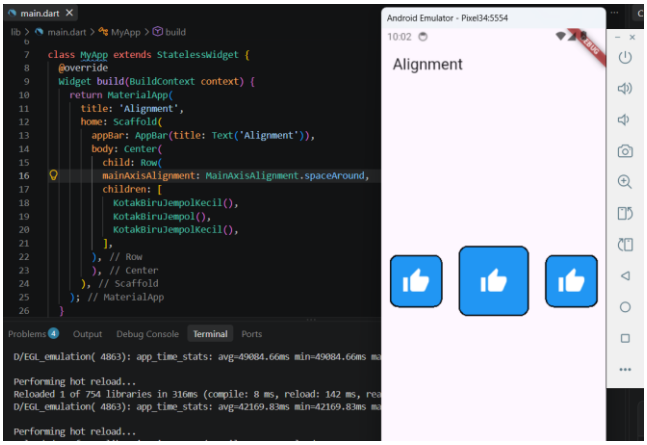
Screenshot bagian kode yang menggunakan:

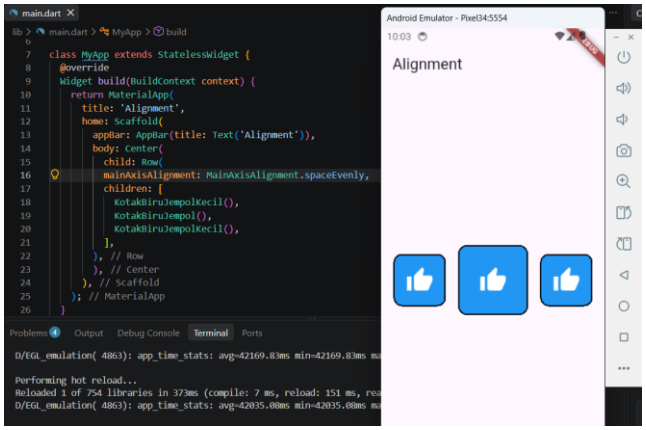


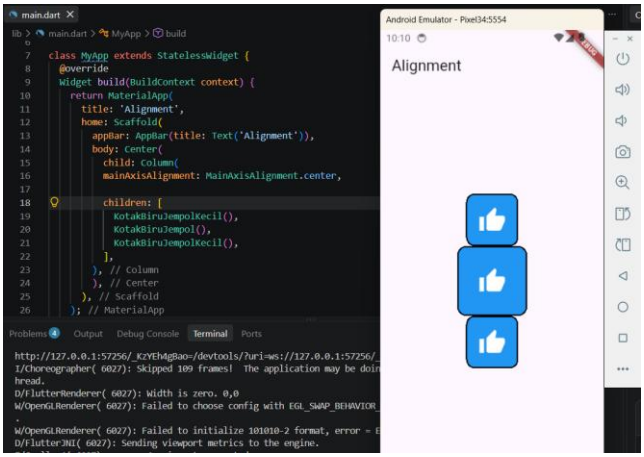

main.dart
lib > main.dart > MyApp > build
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Row(
mainAxisAlignment: MainAxisAlignment.start,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Row
), // Center
), // Scaffold
); // MaterialApp
}
Problems 4 Output Debug Console Terminal Ports
D/EGL_emulation(4863): app_time_stats: avg=110340.51ms min=110340
Performing hot reload...
Reloaded 0 libraries in 104ms (compile: 6 ms, reload: 0 ms, reass
D/EGL_emulation(4863): app_time_stats: avg=6354.23ms min=6354.23
Performing hot reload...
Reloaded 1 of 754 libraries in 335ms (compile: 7 ms, reload: 135
D/EGL_emulation(4863): app_time_stats: avg=28341.50ms min=28341.5
Android Emulator - Pixel34:5554
10:00
Alignment
DEBUG

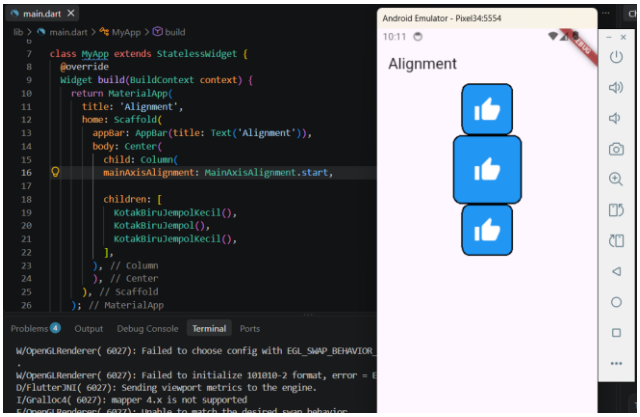

main.dart
lib > main.dart > MyApp > build
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Row(
mainAxisAlignment: MainAxisAlignment.end,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Row
), // Center
), // Scaffold
); // MaterialApp
}
Problems 4 Output Debug Console Terminal Ports
D/EGL_emulation(4863): app_time_stats: avg=6354.23ms min=6354.23
Performing hot reload...
Reloaded 1 of 754 libraries in 335ms (compile: 7 ms, reload: 135
D/EGL_emulation(4863): app_time_stats: avg=28341.50ms min=28341.5
Performing hot reload...
Reloaded 1 of 754 libraries in 285ms (compile: 8 ms, reload: 128
D/EGL_emulation(4863): app_time_stats: avg=49084.66ms min=49084.
Android Emulator - Pixel34:5554
10:00
Alignment
DEBUG

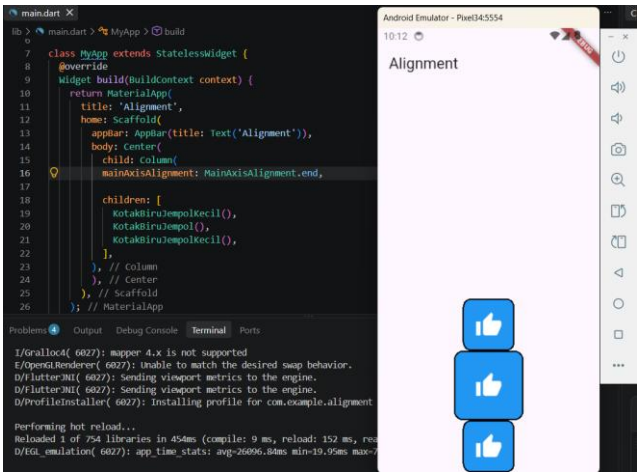

main.dart
lib > main.dart > MyApp > build
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Row(
mainAxisAlignment: MainAxisAlignment.spaceBetween,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Row
), // Center
), // Scaffold
); // MaterialApp
}
Problems 4 Output Debug Console Terminal Ports
D/EGL_emulation(4863): app_time_stats: avg=28341.50ms min=28341.50ms
Performing hot reload...
Reloaded 1 of 754 libraries in 285ms (compile: 8 ms, reload: 128 ms,
D/EGL_emulation(4863): app_time_stats: avg=49084.66ms min=49084.66ms
Performing hot reload...
Android Emulator - Pixel34:5554
10:01
Alignment
DEBUG

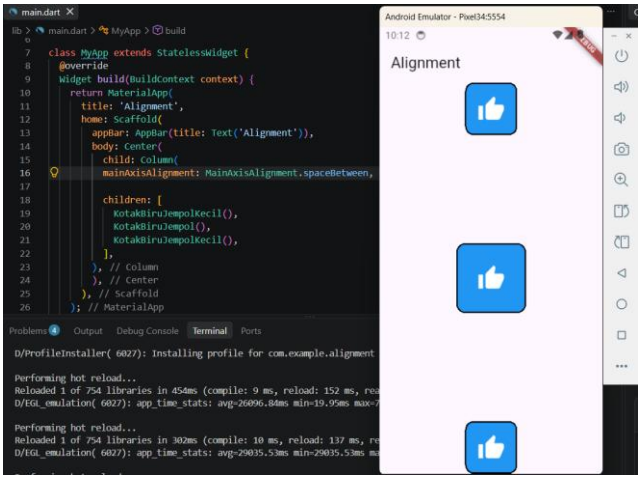

main.dart
lib > main.dart > MyApp > build
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Row(
mainAxisAlignment: MainAxisAlignment.spaceAround,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Row
), // Center
), // Scaffold
); // MaterialApp
}
Problems 4 Output Debug Console Terminal Ports
D/EGL_emulation(4863): app_time_stats: avg=49084.66ms min=49084.66ms
Performing hot reload...
Reloaded 1 of 754 libraries in 316ms (compile: 8 ms, reload: 142 ms,
D/EGL_emulation(4863): app_time_stats: avg=42169.83ms min=42169.83ms
Performing hot reload...
Android Emulator - Pixel34:5554
10:02
Alignment

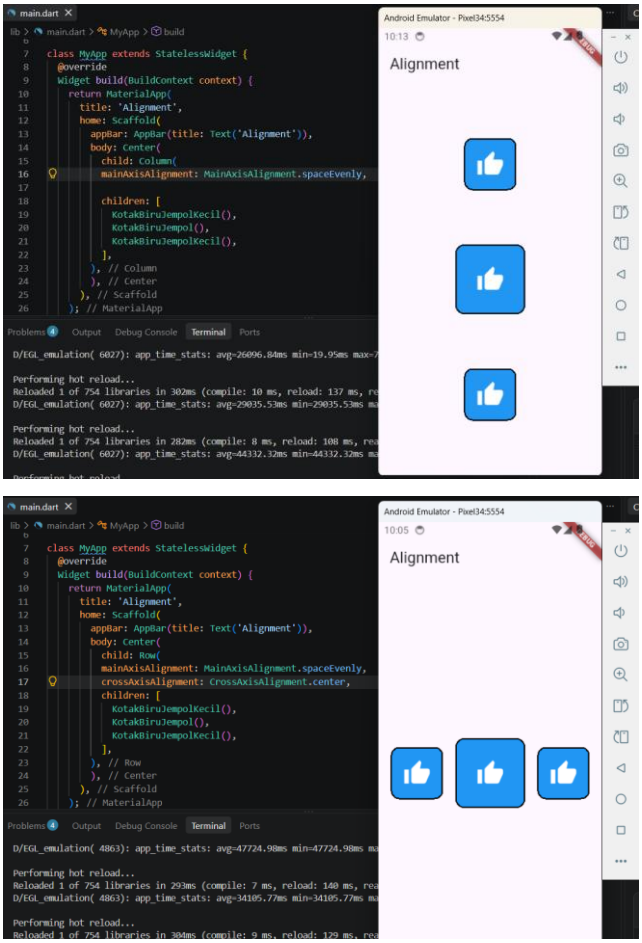

main.dart
lib > main.dart > MyApp > build
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Row(
mainAxisAlignment: MainAxisAlignment.spaceEvenly,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Row
), // Center
), // Scaffold
); // MaterialApp
}
Problems 4 Output Debug Console Terminal Ports
D/EGL_emulation(4863): app_time_stats: avg=42169.83ms min=42169.83ms
Performing hot reload...
Reloaded 1 of 754 libraries in 373ms (compile: 7 ms, reload: 151 ms,
D/EGL_emulation(4863): app_time_stats: avg=42035.08ms min=42035.08ms
Android Emulator - Pixel34:5554
10:03
Alignment


main.dart
lib > main.dart > MyApp > build
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Column(
mainAxisAlignment: MainAxisAlignment.center,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Column
), // Center
), // Scaffold
); // MaterialApp
}
Problems 4 Output Debug Console Terminal Ports
http://127.0.0.1:57256/_KzYEh4gBao=/devtools/?uri=ws://127.0.0.1:57256/
I/Choreographer(6027): Skipped 109 frames! The application may be doi
hread.
D/FlutterRenderer(6027): Width is zero. 0,0
W/OpenGLRenderer(6027): Failed to choose config with EGL_SWAP_BEHAVIOR
.
W/OpenGLRenderer(6027): Failed to initialize 101010-2 format, error =
D/FlutterJNI(6027): Sending viewport metrics to the engine.
Android Emulator - Pixel34:5554
10:10
Alignment


main.dart
lib > main.dart > MyApp > build
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Column(
mainAxisAlignment: MainAxisAlignment.start,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Column
), // Center
), // Scaffold
); // MaterialApp
Problems 4 Output Debug Console Terminal Ports
W/OpenGLRenderer(6027): Failed to choose config with EGL_SWAP_BEHAVIOR_
W/OpenGLRenderer(6027): Failed to initialize 101010-2 format, error =
D/FlutterJNI(6027): Sending viewport metrics to the engine.
I/Gralloc4(6027): mapper 4.x is not supported
Android Emulator - Pixel34:5554
10:11
Alignment


main.dart
lib > main.dart > MyApp > build
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Column(
mainAxisAlignment: MainAxisAlignment.end,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Column
), // Center
), // Scaffold
); // MaterialApp
Problems 4 Output Debug Console Terminal Ports
I/Gralloc4(6027): mapper 4.x is not supported
E/OpenGLRenderer(6027): Unable to match the desired swap behavior.
D/FlutterJNI(6027): Sending viewport metrics to the engine.
D/FlutterJNI(6027): Sending viewport metrics to the engine.
D/ProfileInstaller(6027): Installing profile for com.example.alignment
Performing hot reload...
Reloaded 1 of 754 libraries in 454ms (compile: 9 ms, reload: 152 ms,
D/EGL_emulation(6027): app_time_stats: avg=26096.84ms min=19.95ms
Android Emulator - Pixel34:5554
10:12
Alignment

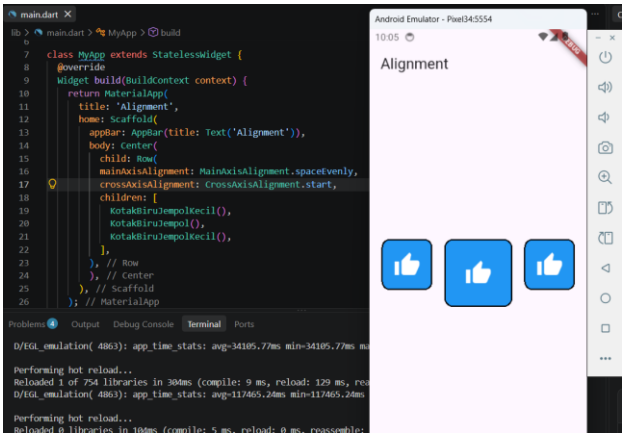

main.dart
lib > main.dart > MyApp > build
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Column(
mainAxisAlignment: MainAxisAlignment.spaceBetween,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Column
), // Center
), // Scaffold
); // MaterialApp
Problems 4 Output Debug Console Terminal Ports
D/ProfileInstaller(6027): Installing profile for com.example.alignment
Performing hot reload...
Reloaded 1 of 754 libraries in 454ms (compile: 9 ms, reload: 152 ms,
D/EGL_emulation(6027): app_time_stats: avg=26096.84ms min=19.95ms
Performing hot reload...
Reloaded 1 of 754 libraries in 302ms (compile: 10 ms, reload: 137 ms,
D/EGL_emulation(6027): app_time_stats: avg=29035.53ms min=29035.53ms
Android Emulator - Pixel34:5554
10:12
Alignment

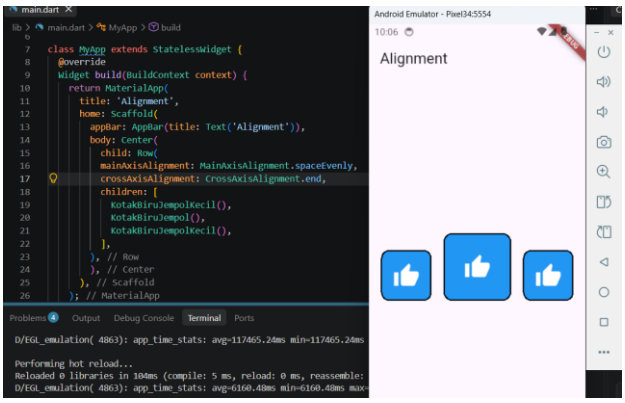


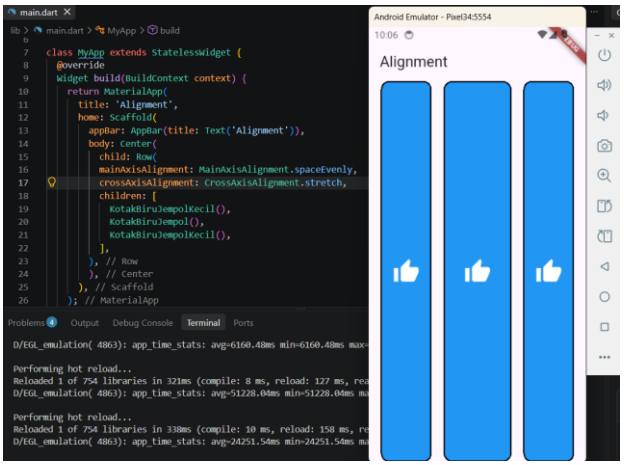
Keterangan : Implementasi berbagai nilai MainAxisAlignment pada Row dan Column untuk mengatur posisi widget pada sumbu utama.

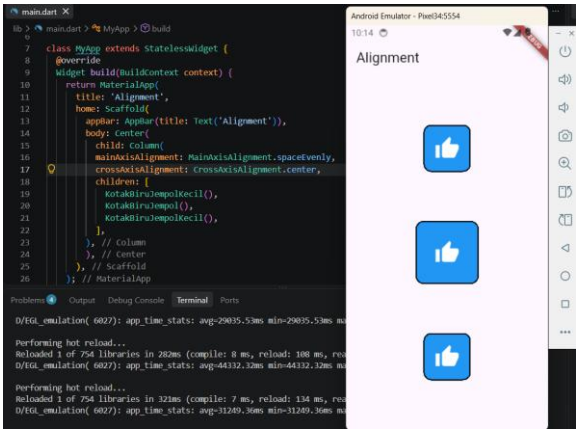
C. Percobaan CrossAxisAlignment

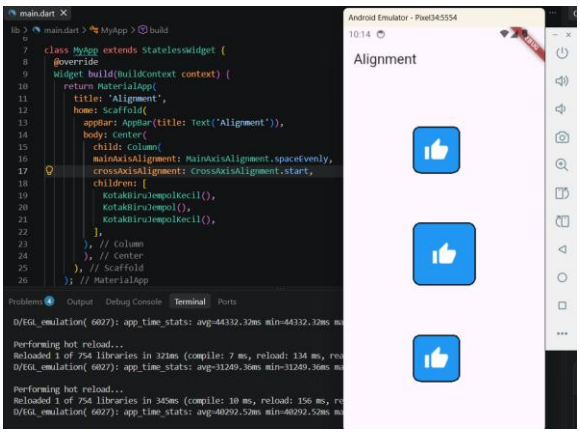
1. Kode Program dan Hasil Percobaan

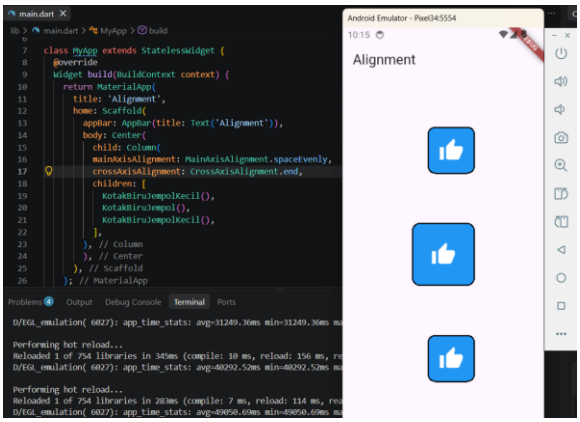


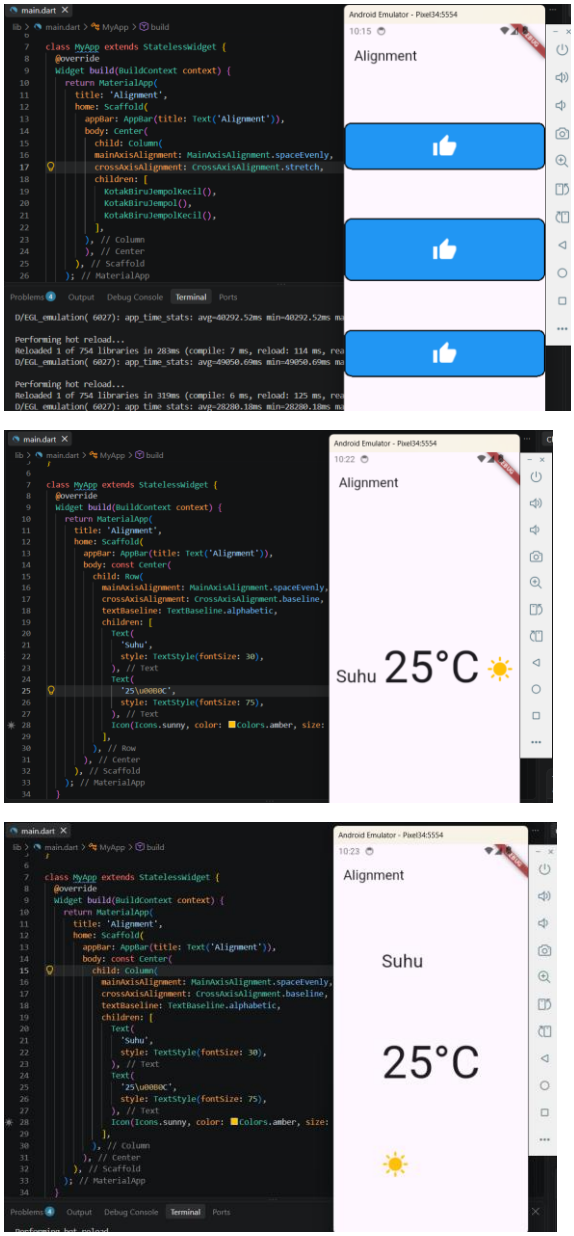





main.dart
Android Emulator - Pixel34:5554
10:14
Alignment
DEBUG
Problems 4
Output
Debug Console
Terminal
Ports
D/EGL_emulation(6027): app_time_stats: avg=29035.53ms min=29035.53ms ma
Performing hot reload...
Reloaded 1 of 754 libraries in 282ms (compile: 8 ms, reload: 108 ms, rea
D/EGL_emulation(6027): app_time_stats: avg=44332.32ms min=44332.32ms ma
Performing hot reload...
Reloaded 1 of 754 libraries in 321ms (compile: 7 ms, reload: 134 ms, rea
D/EGL_emulation(6027): app_time_stats: avg=31249.36ms min=31249.36ms ma


main.dart
Android Emulator - Pixel34:5554
10:14
Alignment
DEBUG
Problems 4
Output
Debug Console
Terminal
Ports
D/EGL_emulation(6027): app_time_stats: avg=44332.32ms min=44332.32ms ma
Performing hot reload...
Reloaded 1 of 754 libraries in 321ms (compile: 7 ms, reload: 134 ms, rea
D/EGL_emulation(6027): app_time_stats: avg=31249.36ms min=31249.36ms ma
Performing hot reload...
Reloaded 1 of 754 libraries in 345ms (compile: 10 ms, reload: 156 ms, re
D/EGL_emulation(6027): app_time_stats: avg=40292.52ms min=40292.52ms ma


main.dart
Android Emulator - Pixel34:5554
10:15
Alignment
DEBUG
Problems 4
Output
Debug Console
Terminal
Ports
D/EGL_emulation(6027): app_time_stats: avg=31249.36ms min=31249.36ms ma
Performing hot reload...
Reloaded 1 of 754 libraries in 345ms (compile: 10 ms, reload: 156 ms, re
D/EGL_emulation(6027): app_time_stats: avg=40292.52ms min=40292.52ms ma
Performing hot reload...
Reloaded 1 of 754 libraries in 283ms (compile: 7 ms, reload: 114 ms, rea
D/EGL_emulation(6027): app_time_stats: avg=49050.69ms min=49050.69ms ma



Gambar 5. Implementasi CrossAxisAlignment untuk mengatur posisi widget pada sumbu silang.

D. Tugas – Tampilan Informasi Cuaca

Nah, ini bagian yang **paling penting** karena merupakan hasil tugas Modul 3.

1. Kode Program



```dart
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Alignment',
      home: Scaffold(
        appBar: AppBar(title: Text('Alignment')),
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            crossAxisAlignment: CrossAxisAlignment.center,
            children: [
              const Text(
                'Malang',
                style: TextStyle(fontSize: 40),
              ),
              const SizedBox(height: 30),
              const Text(
                '25°',
                style: TextStyle(fontSize: 75, fontWeight: FontWeight.bold),
              ),
              const SizedBox(height: 50),
              Row(
                mainAxisAlignment: MainAxisAlignment.spaceEvenly,
                crossAxisAlignment: CrossAxisAlignment.center,
                children: [
                  Column(
                    crossAxisAlignment: CrossAxisAlignment.center,
                    children: const [
                      Text('Minggu', style: TextStyle(fontSize: 16)),
                      SizedBox(height: 10),
                      Icon(Icons.sunny, size: 40),
                      SizedBox(height: 10),
                      Text('20°C', style: TextStyle(fontSize: 16)),
                    ],
                  ),
                  Column(
                    crossAxisAlignment: CrossAxisAlignment.center,
                    children: const [
                      Text('Senin', style: TextStyle(fontSize: 16)),
                      SizedBox(height: 10),
                      Icon(Icons.cloudy_snowing, size: 40),
                      SizedBox(height: 10),
                      Text('23°C', style: TextStyle(fontSize: 16)),
                    ],
                  ),
                  Column(
                    crossAxisAlignment: CrossAxisAlignment.center,
                    children: const [
                      Text('Selasa', style: TextStyle(fontSize: 16)),
                      SizedBox(height: 10),
                      Icon(Icons.cloud, size: 40),
                      SizedBox(height: 10),
                      Text('22°C', style: TextStyle(fontSize: 16)),
                    ],
                  ),
                ],
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

Keterangan Implementasi antarmuka informasi cuaca menggunakan kombinasi Column dan Row serta pengaturan MainAxisAlignment.spaceEvenly.

2. Hasil Tampilan



Keterangan : Hasil implementasi antarmuka informasi cuaca dengan informasi kota, suhu, dan prakiraan cuaca menggunakan beberapa icon cuaca.

IV. Kesimpulan

- Properti mainAxisSize mengendalikan konsumsi luas ruang oleh Row atau Column pada sumbu utamanya, apakah ingin memenuhi layar (*max*) atau hanya membungkus seukuran kontennya (*min*).
- Properti mainAxisAlignment mengendalikan distribusi, perataan letak, dan jarak spasi *widget* anak di sepanjang sumbu utama.
- Properti crossAxisAlignment mengendalikan penempatan posisi *widget* pada sumbu tegak lurus (*cross axis*) terhadap sumbu utama.
- Kombinasi strategis antara Row, Column, dan pengaturan properti *alignment* tersebut sangat vital dalam menyusun desain antarmuka (*layout*) yang rapi, fleksibel, dan responsif terhadap dimensi layar perangkat.